

mode helps customise air-cooler settings as per choice.

In smart mode, if ambient temperature is below 5°C, the unit runs at low wind speed; when ambient temperature is 25°C - 29°C, it runs at medium wind speed; and when ambient temperature is above 29°C, the unit runs at high wind speed. Timer function allows you to set the timer—a feature best for office purposes.

Some air-coolers also feature an air purifying system, like Usha, Symphony and Bajaj, among others. Air ionisers are added to get rid of impurities, dirt, dust and other contaminants from the air drawn into the unit, before it is cooled and discharged.

Child lock function prevents children from changing settings with control buttons.

Low water alarm reminds you about the timely refilling of the air-cooler. When the water in the tank falls below the minimum level marked on the tank, the cooler emits a continuous sound for a few seconds to signal a low water level. Then, cooling mode stops, and the cooler automatically shuts down when the tank is removed.

Some models come with a separate ice chamber to chill the pads. Adding ice is more effective in areas with very high temperatures.

Auto-close shutter feature helps protect the cooler from dust and mosquito-breeding when the unit is not in use. The ability to add mosquito repellent to the cooler is another additional feature—a covered water tank also prevents mosquito breeding.

Auto-drain option allows easy drainage of water.

Remote control function lets you relax and enjoy the cool breeze with the touch of a button.

The processes of water absorption and evaporation can be increased using advanced patterns in the cooling pads. Two popular patterns are honeycomb and aspen (wood wool). Aspen pads are less efficient than honeycomb pads, which are approximately



Evaporative cooling pad (Credit: www.tradekorea.com)

85 per cent efficient.

Honeycomb evaporative cooling pads in an air-cooler are special filters that hold the water for the air to pass through. These pads have good hydrophilic properties that help in absorbing remaining dust particles and cleansing air thoroughly using specially-processed anti-bacterial, anti-erosion and anti-deformation pads.

A special honeycomb is designed to give maximum cooling, low pressure drop and years of use without breaking down. This design is for mixing of air and water for optimum cooling. The unique design directs more water to the front of the pad, where the air is driest, dirtiest and hottest, for better evaporation. The honeycomb is made from an engineered thick cellulose paper, which is chemically treated to resist deterioration.

Benefits of air-coolers

Power consumed by air-coolers is very little as compared to air-conditioners. For example, a 1.5-ton air-con consumes about 1.2 units in an hour, whereas an air-cooler for the same sized room consumes 0.12 units in an hour. The initial cost of buying a cooler is much lower than that of an air-conditioner, which makes the former pocket friendly, too.

Air-coolers are eco-friendly as these do not use any refrigerant for cooling, as used in air-conditioners. These are portable and very easy to place in any room that needs cooling. Their air purifying feature makes the room healthier

for people suffering from allergies, diseases such as asthma and other respiratory ailments, or impaired immunity.

Challenges of air-coolers

In areas with very high humidity, coolers are less efficient. While buying a cooler look for humidity control feature in the specifications if you live in a humid environment.

Air-coolers also need a continuous supply of water.

These also need maintenance more frequently. The fan in the air-coolers makes some noise that could be disturbing. But many recent coolers have designs that minimise the noise.

Tips for the better working of air-coolers

Having good ventilation in a room will work in your cooler's favour. Placing the cooler near a window can be an added advantage—this will cool the room faster with fresh air flowing in. Cleaning and maintaining the cooler timely will help increase its life. Adding ice water or ice in coolers will lower the temperatures even further. So, using ice water is a good idea on very hot days.

These days, many new designs of air-coolers with higher capacities up to 100 litres and other advanced features are available in the market. The range includes both personal and desert coolers, allowing you to choose a cooler of your choice and convenience. Multidirectional wheels provide easy mobility, while quadra flow technology provides uniform cooling. Powerful motors and non-corrosive fan blades increase durability. Low power consumption allows the coolers to be operated on inverter supply in case of a power cut. Features like thermal overload protection, fragrance chamber, LCD display, auto-drain for easy drainage of water, mosquito repellents and ionisers to purify air make such multipurpose smart coolers the best choice. **EFY**